DAY 05 | HANDS-ON ACTIVITIES

API Fundamentals and FastAPI: Problem Statements

Six problems. Use TestClient in Colab, or run uvicorn and use /docs in VS Code.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Design the Endpoints

Level: Warm-up | **Concepts:** REST design, methods, status codes

For a library app with books and members, list 8 endpoints with method, URL, purpose and success status code. No code needed.

Sample run / expected result

```
GET    /books           list books       200
POST   /books           add a book       201
DELETE /books/{id}      remove a book    204
```

Hints

- Use plural nouns and no verbs in URLs.

Problem 2: Hello API

Level: Easy | **Concepts:** FastAPI setup, GET

Create a FastAPI app with GET / returning a welcome message and GET /health returning {"status": "ok"}. Run it and open /docs.

Sample run / expected result

```
GET /health -> 200 {"status": "ok"}
```

Hints

- uvicorn main:app --reload
- In Colab: TestClient(app).get("/health").json()

Problem 3: Path Parameters

Level: Easy | **Concepts:** path params, type hints

Add GET /square/{n} that returns n and n squared. What happens if you call /square/abc? Note the status code.

Sample run / expected result

```
GET /square/7 -> {"n": 7, "square": 49}
GET /square/abc -> 422
```

Hints

- Declare n: int and FastAPI validates for you.

Problem 4: Quotes API

Level: Medium | **Concepts:** GET list, GET by id, 404

Store 5 quotes in a dict. Add GET /quotes, GET /quotes/{id} (404 if missing), and GET /quotes/random.

Sample run / expected result

```
GET /quotes/99 -> 404 {"detail": "Quote not found"}
```

Hints

- raise HTTPException(status_code=404, detail="Quote not found")
- import random; random.choice(list(...))

Problem 5: Create and Delete

Level: Challenging | **Concepts:** POST 201, DELETE 204

Add POST /quotes that accepts text and author as query parameters and returns 201 with the new id. Add DELETE /quotes/{id} returning 204.

Sample run / expected result

```
POST /quotes?text=Keep going&author=Anon -> 201 {"id": 6, ...}
DELETE /quotes/6 -> 204
```

Hints

- @app.delete("/quotes/{qid}", status_code=204)

Problem 6: Test Everything

Level: Stretch | **Concepts:** TestClient, assertions

Write at least 8 assert based tests using TestClient that check status codes and response bodies for all your endpoints, including error cases.

Sample run / expected result

```
assert client.get("/quotes/99").status_code == 404
All 8 tests passed
```

Hints

- from fastapi.testclient import TestClient

Ganbatte (頑張って): do your best